

Software Requirements for Math Madness

Math Madness

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Overview

This document describes the requirements for the software implementation of the classic Memory game, with a math twist. Following this software will allow 2 players to play the game of memory together.

The Game

The players take turns choosing two “cards”, one from each section. When a card from the second section presents the answer to the example on the first card, the match is taken and the player earns a point. Otherwise, when it isn’t a match, the cards get returned to their original positions.

Software Implementation

UI Elements

The software will present the memory game with the following UI elements:

- 2 sections with 8 cards in each
- Start button
- Label to display game status and if correct or not
- Button to change player turn
- Label to display each player’s points

Game Process and Rules

- The game starts when the player clicks Start. Until then all clicks on the spot buttons are ignored, and the game status message is “Click Start to begin Game.”
- After Start is clicked: The message “Player 1” is displayed. Player 1 takes a turn by clicking on a question card in section 1-and the card displays it’s question, and an answer card in section 2. If the answer matches the question, player 1 gets a point. Otherwise, the cards are returned to original position. Click new turn and it is now player 2’s turn and the game continues.
- When correct match is chosen the cards turn green and the status shows correct, when it is incorrect the cards turn red and the status shows incorrect.
- When all matches are taken, the game is over. The label will display which player is the winner or if it is a tie.
- When the round finishes the cards should randomly switch places, to create a new board.
- If the user clicks Start in middle of a game, then the game should interrupt and begin a new game.